# LAB EXPERIMENT RECORD

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|---|---|
| **Department** | AIDS |
| **Course Code / Course Name** | DSA0601 / Data Handling and Visualization |
| **Experiment** | Data Visualization using R |
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| **Register Number** | 192424055 |
| **Date** | 15 October 2026 |
| **Day** | Thursday |

# SET 22: Time Series Analysis

### Aim

To analyze monthly sales data using a time-series line chart, advertising-versus-sales scatter plot, Tableau dashboard, and autocorrelation plot.

### Dataset: Monthly Sales Data

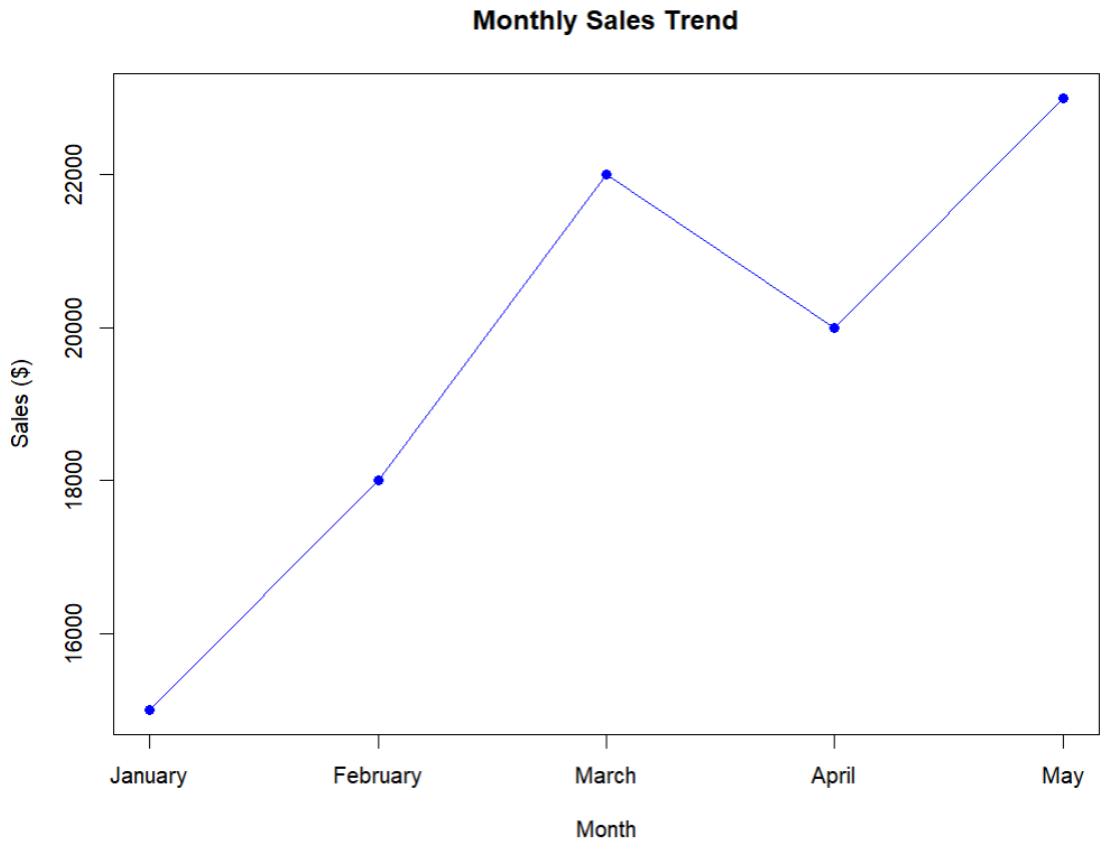
| Month | Sales ($) |
|---|---|
| January | 15000 |
| February | 18000 |
| March | 22000 |
| April | 20000 |
| May | 23000 |

**Q1. In R, create a time-series line chart to visualize the trend in monthly sales.**

**R Program**

```
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(
  sales,
  type = "o",
  pch = 19,
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales ($)",
  main = "Monthly Sales Trend",
  col = "blue"
)
axis(1, at = 1:5, labels = month)
```

**Output:**




**Observation:** Sales rise from January to March, fall slightly in April, and reach their highest value in May at $23,000.

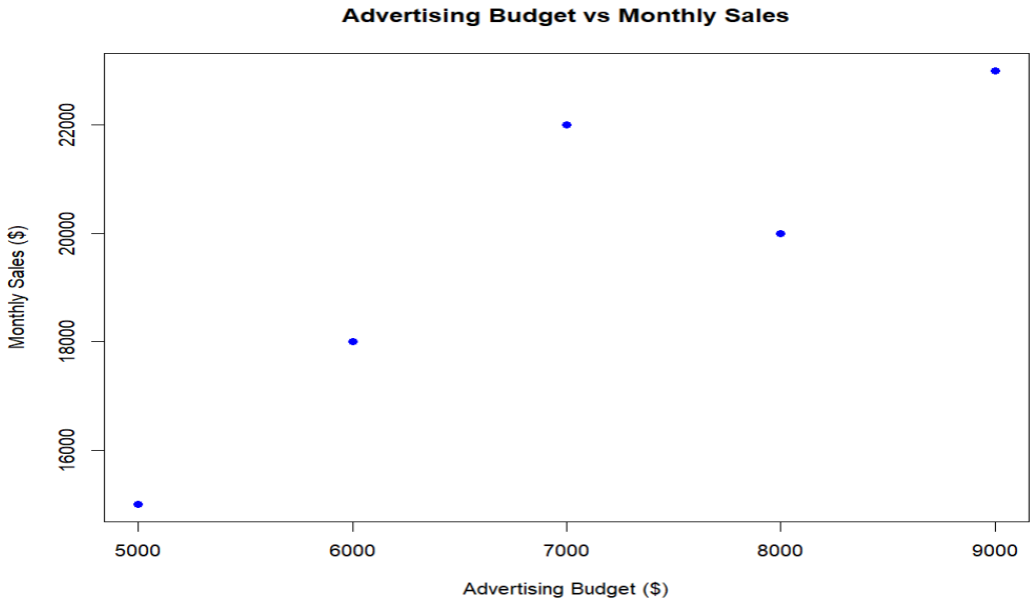
**Q2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales.**

| Month | Advertising Budget ($) | Sales ($) |
|---|---|---|
| January | 5000 | 15000 |
| February | 6000 | 18000 |
| March | 7000 | 22000 |
| April | 8000 | 20000 |
| May | 9000 | 23000 |

**R Program**

```
advertising <- c(5000, 6000, 7000, 8000, 9000)
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(
  advertising,
  sales,
  pch = 19,
  col = "blue",
  xlab = "Advertising Budget ()", ylab = "MonthlySales()",
  main = "Advertising Budget vs Monthly Sales"
)
correlation <- cor(advertising, sales)
print(correlation)
```

**Output:**



## Q3. In Tableau, build a dashboard combining a line chart and a pie chart.

Step 1 — Prepare the CSV files

Save the monthly data as Monthly_Sales.csv:

Month,Sales
January,15000
February,18000
March,22000
April,20000
May,23000

For the pie chart, save a separate file as Product_Sales.csv:

Product,Sales
Product A,30000
Product B,25000
Product C,18000

These product figures are sample values, not values from the supplied dataset.

Step 2 — Create the monthly sales line chart

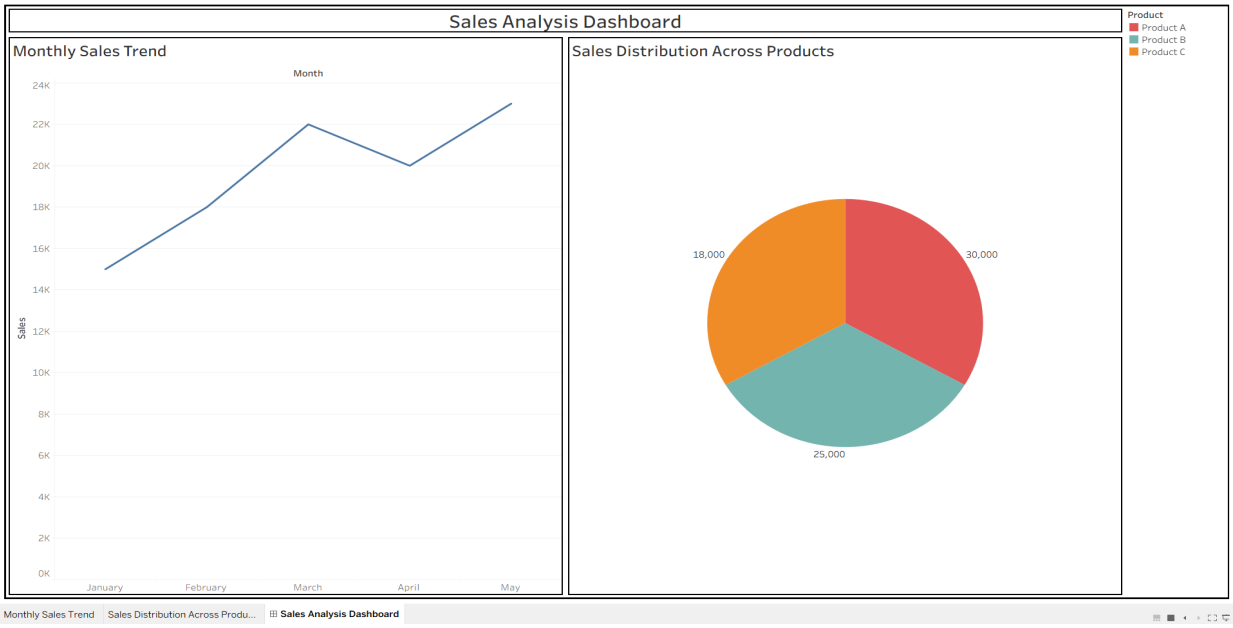
1. Open Tableau and connect to Monthly_Sales.csv.
2. Create a new worksheet named Monthly Sales Trend.
3. Drag Month → Columns.
4. Drag Sales → Rows.
5. Select Line from the Marks dropdown.
6. Arrange the months in calendar order: January, February, March, April, May.
7. Set the title to Monthly Sales Trend.

Step 3 — Create the product sales pie chart

1. Connect to Product_Sales.csv as another data source.
2. Create a new worksheet named Product Sales Distribution.
3. Select Pie from the Marks dropdown.
4. Drag Product → Color.
5. Drag Sales → Angle.
6. Drag Sales → Label.
7. Set the title to Sales Distribution Across Products.

Step 4 — Create the dashboard

1. Click New Dashboard.
2. Add Monthly Sales Trend.
3. Add Product Sales Distribution.
4. Arrange the line chart and pie chart side by side.
5. Add suitable filters if required.
6. Set the dashboard title to Sales Analysis Dashboard.

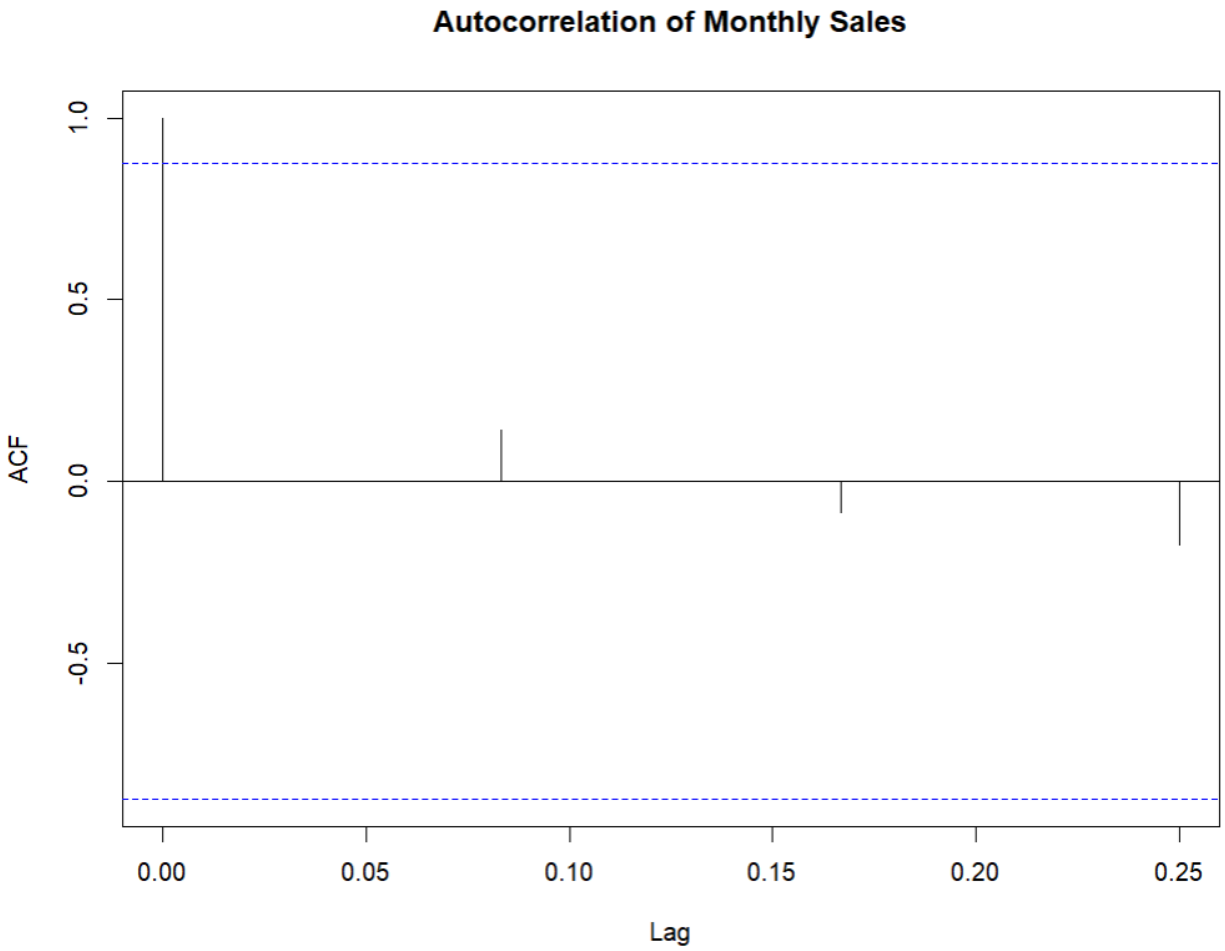



**Q4. In R, create an autocorrelation plot to identify seasonality in the time series data.**

**R Program**

```
sales <- c(15000, 18000, 22000, 20000, 23000)
sales_ts <- ts(sales, start = c(2023, 1), frequency = 12)
acf(
  sales_ts,
  main = "Autocorrelation of Monthly Sales",
  lag.max = 3
)
```

**Output:**




**Observation:** The ACF plot helps examine whether sales values are related to earlier months. However, the dataset contains only five months, which is too short to identify reliable seasonal patterns. At least several complete yearly cycles of monthly data would be preferable for studying annual seasonality.

## Procedure

1. Imported the monthly sales data into R.
2. Created a time-series line chart to visualize sales trends.
3. Generated a scatter plot to compare advertising budget and sales using illustrative advertising data.
4. Created a Tableau dashboard containing a sales trend line chart and product sales pie chart using illustrative product data.
5. Generated an autocorrelation plot to examine relationships between sales values at different lags.

## Result

Monthly sales trends were visualized using R, and the relationship between advertising budget and sales was explored using sample data.